

5.

(a) In logistic regression, for each data point,

$$t_n \in \{0, 1\}$$

WKT//

$$p(t_n = 1 | x_n, w) = \sigma(w^T x_n) = \frac{1}{1 + e^{-w^T x_n}}$$

$$\text{Let } y_n = \sigma(w^T x_n)$$

$$\Rightarrow p(t_n = 1 | x_n, w) = y_n$$

$$\therefore p(t_n = 0 | x_n, w) = 1 - y_n //$$

For a pt,

$$p(t_n | x_n, w) = y_n^{t_n} (1 - y_n)^{1 - t_n}$$

For N observations,

$$p(t | X, w) = \prod_{n=1}^N y_n^{t_n} (1 - y_n)^{1 - t_n}$$

Taking log.

$$= \sum_{n=1}^N [t_n \log y_n + (1 - t_n) \log (1 - y_n)]$$

$$E(w) = - \sum_{n=1}^N [t_n \log y_n + (1 - t_n) \log (1 - y_n)]$$

Taking the gradient,

$$\begin{aligned} \nabla E(w) &= \sum_{n=1}^N (y_n - t_n) x_n \\ &= \sum_{n=1}^N [\sigma(w^T x_n) - t_n] x_n // \end{aligned}$$

$$\text{Let } \phi = [x_1 \quad x_2 \quad \dots \quad x_N], \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \quad \& \quad t = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix}$$

$$\Rightarrow \nabla E(w) = \phi(y - t)$$

Differentiating gradient again,

at each point,

$$\frac{\partial y_n}{\partial w} = y_n(1-y_n)x_n$$

$$\text{Let } r_n = y_n(1-y_n)$$

$$\& \quad R = \begin{bmatrix} r_1 & 0 & \dots & 0 \\ 0 & r_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_N \end{bmatrix}$$

$$\therefore \text{Hessian, } H = \nabla^2 E(w) = \Phi R \Phi^T$$

Newton-Raphson method to minimize $E(w)$:

$$w_{\text{new}} = w_{\text{old}} - H^{-1} \nabla E$$

Substituting the values,

$$w_{\text{new}} = w_{\text{old}} - (\Phi R \Phi^T)^{-1} \Phi (y - t)$$

$$w_{\text{new}} = w_{\text{old}} + (\Phi R \Phi^T)^{-1} \Phi (t - y) //$$

Algorithm

Step 1: Initialize : Choose $w^{(0)}$

Step 2: Calculate predictions, for each n ,

$$y_n^{(k)} = \sigma((w^{(k)})^T x_n)$$

Step 3: Construct R

$$r_n^{(k)} = y_n^{(k)}(1 - y_n^{(k)})$$

$$R^{(k)} = \text{diag}(r_1^{(k)}, r_2^{(k)}, \dots, r_N^{(k)})$$

Step 4: Calculate Gradient:

$$\nabla E(w^{(k)}) = \Phi (y^{(k)} - t)$$

Step 5: Calculate Hessian

$$H^{(k)} = \Phi R^{(k)} \Phi^T$$

Step 6: Update weights

$$\begin{aligned} \omega^{(k+1)} &= \omega^{(k)} - (H^{(k)})^{-1} \nabla E(\omega^{(k)}) \\ &= \omega^{(k)} + (\Phi R^{(k)} \Phi^T)^{-1} \Phi (t - y^{(k)}) \end{aligned}$$

Step 7: Repeat until convergence (change in ω or $E(\omega) < \text{threshold}$)

Step 8: Return ω

(b) WKTH

$$\begin{aligned} \omega_{\text{new}} &= \omega_{\text{old}} + (\Phi R \Phi^T)^{-1} \Phi^T (t - y) \\ &= (\Phi R \Phi^T)^{-1} [\omega (\Phi R \Phi^T) \omega_{\text{old}} + \Phi^T (t - y)] \\ &= (\Phi R \Phi^T)^{-1} \Phi R [\Phi^T \omega_{\text{old}} + R^{-1} (t - y)] \end{aligned}$$

$$\text{Let } z = \Phi^T \omega_{\text{old}} + R^{-1} (t - y)$$

$$\Rightarrow \omega_{\text{new}} = (\Phi R \Phi^T)^{-1} \Phi R z$$

which is exactly the weighted least squares solution from 4(c) ~~was~~ except weights $r_n = y_n \cdot (1 - y_n)$ and target t_n replaced by pseudo targets z_n

Unlike the std weighted least squares, the matrix R & effective target z both depend on ω_{old} through predictions y .

$\therefore R$ & z must be re-evaluated at every iteration based on ω_{new} , the optimization technique is named Iteratively Reweighted Least Squares

(c)

WKT//

$$H = \Phi R \Phi^T \quad \text{where } r_n = y_n(1-y_n)$$

$$\because 0 < y_n < 1 \Rightarrow y_n(1-y_n) > 0$$

$$\therefore R \geq 0 \quad [R \text{ is +ve semi definite}]$$

Consider an arbitrary vector a ,

$$\begin{aligned} a^T H a &= a^T \Phi R \Phi^T a \\ &= (\Phi^T a)^T R (\Phi^T a) \end{aligned}$$

$\because R$ is positive semi-definite,

$$(\Phi^T a)^T R (\Phi^T a) \geq 0$$

$$\therefore H \geq 0$$

$\therefore E(w)$ is convex

If H is positive definite, then $E(w)$ is strictly convex

$\Rightarrow$ there can only be one global minimum

$$\therefore H = \Phi R \Phi^T$$

$\Rightarrow H$ is positive definite when Φ has full row rank

$\therefore$ if Φ has full row rank

$$\Rightarrow H \succ 0$$

$\Rightarrow E(w)$ is strictly convex

$\Rightarrow$ unique global minimum//